Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Brasil & Sons Dairy # 2

Physical address of dairy:

Number and Street	City	County	Zip Code
21728 Lone Tree RD	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X205-X230-XX08-XXXX X205-X240-XX19-XXXX

B. OPERATORS

Brasil, George

Operator name: Brasil, George

Telephone no.: Landline: ; Cellular: (209) 595-3903

Mailing Address Number and Street	City	State	Zip Code
21728 Lone Tree RD	Escalon	CA	95320

This operator is responsible for paying permit fees.

Brasil, Victor

Operator name: Brasil, Victor

Telephone no.: Landline: ; Cellular: (209) 595-1577

Mailing Address Number and Street	City	State	Zip Code
21728 Lone Tree RD	Escalon	CA	95320

C. OWNERS

Azevedo, Art

Legal owner name: Azevedo, Art

Telephone no.: Landline: (209) 526-3560; Cellular: (209) 985-3560

Mailing Address Number and Street	City	State	Zip Code
1026 Harvard AVE	Modesto	CA	95350

Azevedo, David

Legal owner name:	Azevedo, David	Telephone no.:	Landline	(209) 204-8492 Cellular
20350 Allen RD	Escalon	CA	95320	
Mailing Address Number and Street	City	State	Zip Code	

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	120	125	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	120	125	0	0
Average number	0	0	120	125	0	0
Avg live weight (lbs)	0	0	1,000	750		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 2,458.47 *tons per reporting period*

Total nitrogen from manure: 23,250.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 16,275.35 *lbs per reporting period*

Total phosphorus from manure: 3,934.70 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: _____ *gallons*

Total nitrogen generated: _____ *lbs*

Total phosphorus generated: _____ *lbs*

Total potassium generated: _____ *lbs*

Total salt generated: _____ *lbs*

	0	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	0	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Creek	Surface water
SSJID Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	Corral solids *Corral Solids*	1,070.00 *ton*	Dry-weight	57.0	30,400.00	15,100.00	49,100.00		25.80

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	27,974.08	13,895.02	45,181.82	237,411.60
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	27,974.08	13,895.02	45,181.82	237,411.60

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Brennan/120	25	20	2	manure	X247-X020-XX07-XXXX
D2-1	37	37	1	none	X205-X240-XX19-XXXX
D2-2	14	14	2	manure	X205-X240-XX19-XXXX
D2-3	42	42	2	manure	X205-X240-XX19-XXXX
D2-4	42	42	2	manure	X205-X240-XX19-XXXX
D2-5	15	15	2	manure	X205-X240-XX19-XXXX
Raymus Manteca	27	27	1	none	X218-X020-XX15-XXXX X218-X080-XX03-XXXX X218-X090-XX04-XXXX
Totals for areas that were used for application	138	133	10		
Totals for areas that were not used for application	64	64	2		
Land application area totals	202	197	12		

B. CROPS AND HARVESTS

Brennan/120

Field name: Brennan/120

11/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	330.00 *ton*	As-is		72.2	3,860.00	670.00	4,730.00		8.65

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.50	127.38	22.11	156.09	793.55

Brennan/120

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	536.00 *ton*	As-is		67.2	3,740.00	590.00	3,230.00		4.84

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.80	200.46	31.62	173.13	850.91

D2-1

Field name: D2-1

10/20/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 37 Plant date: 10/20/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/22/2023	290.20 *ton*	As-is		10.0	18,400.00	2,620.00	18,600.00		8.29

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.84	288.63	41.10	291.77	1,170.37

D2-2

Field name: D2-2

D2-2

11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 14 Plant date: 11/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	189.40 *ton*	As-is		60.6	5,480.00	960.00	7,320.00		9.54

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.53	148.27	25.97	198.06	1,017.01

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	368.20 *ton*	As-is		64.0	3,850.00	600.00	3,460.00		4.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.30	202.51	31.56	182.00	833.18

D2-3

Field name: D2-3

11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 42 Plant date: 11/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	648.20 *ton*	As-is		61.6	4,880.00	880.00	5,820.00		9.29

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.43	150.63	27.16	179.64	1,101.13

D2-3

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 42 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	1,121.40 *ton*	As-is		67.2	3,050.00	560.00	3,380.00		5.76

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.70	162.87	29.90	180.49	1,008.88

D2-4

Field name: D2-4

11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 42 Plant date: 11/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	636.30 *ton*	As-is		60.2	4,780.00	990.00	6,740.00		9.86

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.15	144.83	30.00	204.22	1,189.06

06/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 42 Plant date: 06/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,146.60 *ton*	As-is		72.4	3,040.00	600.00	3,540.00		6.37

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	27.30	165.98	32.76	193.28	959.93

D2-5

Field name: D2-5

11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 11/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	209.50 *ton*	As-is		60.6	5,480.00	960.00	7,320.00		9.54

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.97	153.07	26.82	204.47	1,049.95

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	394.50 *ton*	As-is		64.0	3,850.00	600.00	3,460.00		4.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.30	202.51	31.56	182.00	833.18

Raymus Manteca

Field name: Raymus Manteca

10/30/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 27 Plant date: 10/30/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/22/2023	199.80 *ton*	As-is		10.0	25,900.00	2,580.00	23,710.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.40	383.32	38.18	350.91	1,571.76

NUTRIENT BUDGET

A. LAND APPLICATIONS

Brennan/120 - 11/22/2022: Oats, silage-soft dough

Field name: Brennan/120

Crop: Oats, silage-soft dough — Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	0.00	2,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	126.10	44.40	210.01	1,474.08	150.00 *ton*
Application event totals		126.10	44.40	210.01	1,474.08	

Brennan/120 - 06/05/2023: Corn, silage

Field name: Brennan/120

Crop: Corn, silage — Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	261.44	129.86	422.26	2,218.80	200.00 *ton*
Application event totals		261.44	129.86	422.26	2,218.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	65.34	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

Brennan/120 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	59.90	2,475,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	54.45	2,250,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	58.08	2,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	56.27	2,325,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	54.45	2,250,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	56.27	2,325,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.27	

Brennan/120 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	50.82	2,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.82	

D2-1 - 10/20/2019: Alfalfa, hay

Field name: D2-1

Crop: Alfalfa, hay

Plant date: 10/20/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	25.44	0.00	0.00	525.65	4,661,280.00 *gal*
Application event totals		25.44	0.00	0.00	525.65	

04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	25.20	0.00	0.00	520.60	4,616,460.00 *gal*
Application event totals		25.20	0.00	0.00	520.60	

05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	24.46	0.00	0.00	505.44	4,482,000.00 *gal*
Application event totals		24.46	0.00	0.00	505.44	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	24.95	0.00	0.00	515.55	4,571,640.00 *gal*
Application event totals		24.95	0.00	0.00	515.55	

D2-1 - 10/20/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	25.93	0.00	0.00	535.76	4,750,920.00 *gal*
Application event totals		25.93	0.00	0.00	535.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	26.18	0.00	0.00	540.82	4,795,740.00 *gal*
Application event totals		26.18	0.00	0.00	540.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	25.44	0.00	0.00	525.65	4,661,280.00 *gal*
Application event totals		25.44	0.00	0.00	525.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	24.22	0.00	0.00	500.38	4,437,180.00 *gal*
Application event totals		24.22	0.00	0.00	500.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	23.97	0.00	0.00	495.33	4,392,360.00 *gal*
Application event totals		23.97	0.00	0.00	495.33	

D2-2 - 11/23/2022: Oats, silage-soft dough

Field name: D2-2

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

D2-2 - 11/23/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	120.09	42.29	200.01	1,403.89	100.00 *ton*
Application event totals		120.09	42.29	200.01	1,403.89	

11/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	11.25	0.00	0.00	76.94	1,613,520.00 *gal*
Application event totals		11.25	0.00	0.00	76.94	

D2-2 - 06/08/2023: Corn, silage

Field name: D2-2

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	205.42	102.03	331.78	1,743.34	110.00 *ton*
Application event totals		205.42	102.03	331.78	1,743.34	

06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	10.34	0.00	0.00	213.73	717,120.00 *gal*
SSJID Canal	Surface water	0.00	0.00	0.00	41.49	1,200,000.00 *gal*
Application event totals		10.34	0.00	0.00	255.21	

D2-2 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	9.05	0.00	0.00	187.01	627,480.00 *gal*
SSJID Canal	Surface water	0.00	0.00	0.00	36.30	1,050,000.00 *gal*
Application event totals		9.05	0.00	0.00	223.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	57.04	1,650,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	51.86	1,500,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	54.45	1,575,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	49.27	1,425,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	51.86	1,500,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.86	

D2-2 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	49.27	1,425,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.27	

D2-3 - 11/23/2022: Oats, silage-soft dough

Field name: D2-3

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	120.09	42.29	200.01	1,403.89	300.00 *ton*
Application event totals		120.09	42.29	200.01	1,403.89	

11/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	11.46	0.00	0.00	78.37	4,930,200.00 *gal*
Application event totals		11.46	0.00	0.00	78.37	

D2-3 - 06/08/2023: Corn, silage

Field name: D2-3

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

D2-3 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	199.19	98.94	321.72	1,690.51	320.00 *ton*
Application event totals		199.19	98.94	321.72	1,690.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	9.91	0.00	0.00	204.82	2,061,720.00 *gal*
SSJID Canal	Surface water	0.00	0.00	0.00	39.76	3,450,000.00 *gal*
Application event totals		9.91	0.00	0.00	244.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	9.48	0.00	0.00	195.92	1,972,080.00 *gal*
SSJID Canal	Surface water	0.00	0.00	0.00	38.03	3,300,000.00 *gal*
Application event totals		9.48	0.00	0.00	233.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	55.32	4,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	51.86	4,500,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	53.59	4,650,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.59	

D2-3 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	54.45	4,725,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	51.86	4,500,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	50.13	4,350,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.13	

D2-4 - 11/23/2022: Oats, silage-soft dough

Field name: D2-4

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	120.09	42.29	200.01	1,403.89	300.00 *ton*
Application event totals		120.09	42.29	200.01	1,403.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	11.46	0.00	0.00	78.37	4,930,200.00 *gal*
Application event totals		11.46	0.00	0.00	78.37	

D2-4 - 06/06/2023: Corn, silage

Field name: D2-4

Crop: Corn, silage

Plant date: 06/06/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	199.19	98.94	321.72	1,690.51	320.00 *ton*
Application event totals		199.19	98.94	321.72	1,690.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	10.34	0.00	0.00	213.73	2,151,360.00 *gal*
SSJID Canal	Surface water	0.00	0.00	0.00	41.49	3,600,000.00 *gal*
Application event totals		10.34	0.00	0.00	255.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	9.48	0.00	0.00	195.92	1,972,080.00 *gal*
SSJID Canal	Surface water	0.00	0.00	0.00	38.03	3,300,000.00 *gal*
Application event totals		9.48	0.00	0.00	233.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	55.32	4,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	53.59	4,650,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.59	

D2-4 - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	51.86	4,500,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	50.99	4,425,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	50.99	4,425,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	50.13	4,350,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.13	

D2-5 - 11/23/2022: Oats, silage-soft dough

Field name: D2-5

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	123.29	43.41	205.35	1,441.32	110.00 *ton*
Application event totals		123.29	43.41	205.35	1,441.32	

D2-5 - 11/23/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	11.09	0.00	0.00	75.80	1,703,160.00 *gal*
Application event totals		11.09	0.00	0.00	75.80	

D2-5 - 06/08/2023: Corn, silage

Field name: D2-5

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	209.15	103.89	337.81	1,775.04	120.00 *ton*
Application event totals		209.15	103.89	337.81	1,775.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	10.26	0.00	0.00	211.95	761,940.00 *gal*
SSJID Canal	Surface water	0.00	0.00	0.00	41.14	1,275,000.00 *gal*
Application event totals		10.26	0.00	0.00	253.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	9.05	0.00	0.00	187.01	672,300.00 *gal*
SSJID Canal	Surface water	0.00	0.00	0.00	36.30	1,125,000.00 *gal*
Application event totals		9.05	0.00	0.00	223.31	

D2-5 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	58.08	1,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.08	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	53.24	1,650,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.24	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	55.66	1,725,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.66	

08/28/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	48.40	1,500,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.40	

09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	50.82	1,575,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.82	

09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	48.40	1,500,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.40	

Raymus Manteca - 10/30/2019: Alfalfa, hay

Raymus Manteca - 10/30/2019: Alfalfa, hay

Field name: Raymus Manteca

Crop: Alfalfa, hay

Plant date: 10/30/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	59.16	3,300,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	53.78	3,000,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	59.16	3,300,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	51.09	2,850,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	61.85	3,450,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	59.16	3,300,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.16	

Raymus Manteca - 10/30/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	61.85	3,450,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	61.85	3,450,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.85	

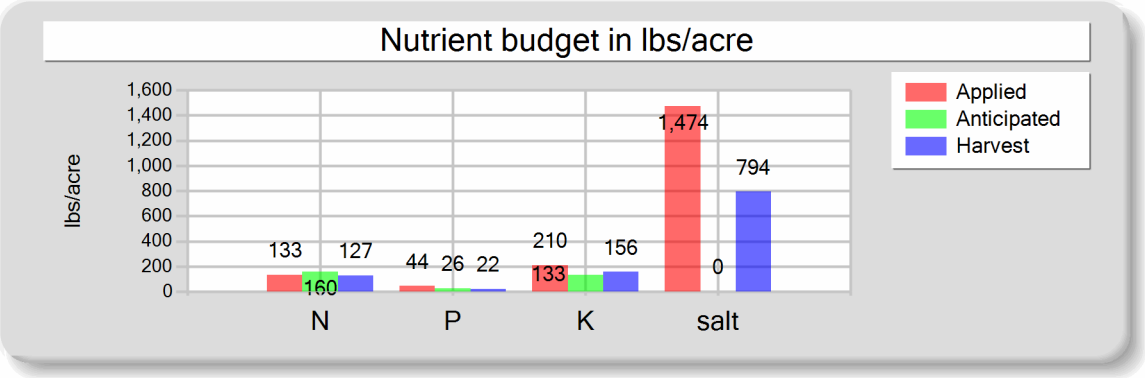
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SSJID Canal	Surface water	0.00	0.00	0.00	56.47	3,150,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.47	

B. NUTRIENT BUDGET

Brennan/120 - 11/22/2022: Oats, silage-soft dough

Field name: Brennan/120 Crop: Oats, silage-soft dough Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	126.10	44.40	210.01	1,474.08
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	133.10	44.40	210.01	1,474.08
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	127.38	22.11	156.09	793.55
Nutrient balance	5.72	22.29	53.92	680.53
Applied to removed ratio	1.04	2.01	1.35	1.86

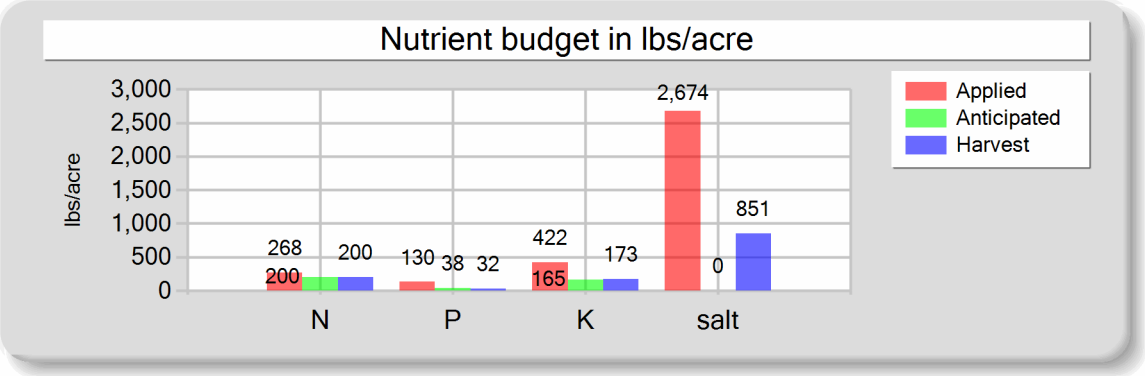
Fresh water applied	
2,400,000.00	*gallons*
88.38	*acre-inches*
4.42	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Brennan/120 - 06/05/2023: Corn, silage

Field name: Brennan/120 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	261.44	129.86	422.26	2,218.80
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	455.57
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	268.44	129.86	422.26	2,674.37
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	200.46	31.62	173.13	850.91
Nutrient balance	67.98	98.24	249.13	1,823.46
Applied to removed ratio	1.34	4.11	2.44	3.14

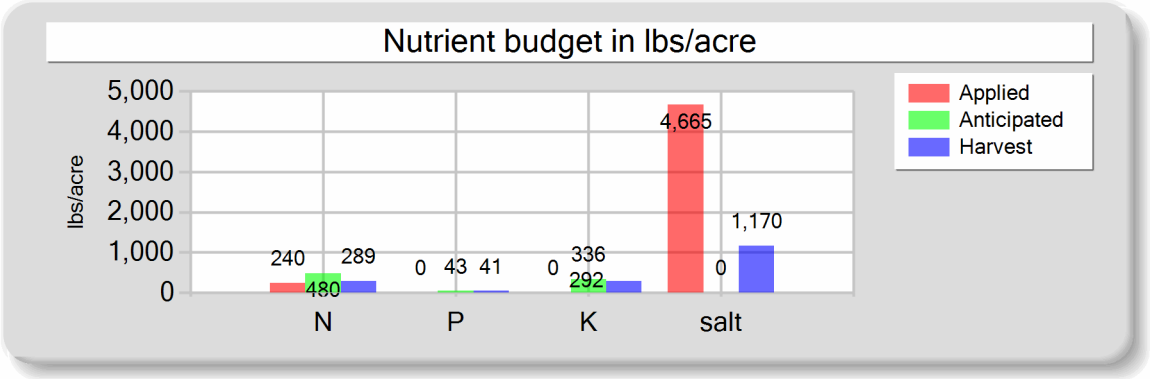
Fresh water applied	
18,825,000.00	*gallons*
693.26	*acre-inches*
34.66	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

D2-1 - 10/20/2019: Alfalfa, hay

Field name: D2-1 Crop: Alfalfa, hay Plant date: 10/20/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	225.79	0.00	0.00	4,665.18
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	239.79	0.00	0.00	4,665.18
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	288.63	41.10	291.77	1,170.37
Nutrient balance	-48.84	-41.10	-291.77	3,494.81
Applied to removed ratio	0.83	0.00	0.00	3.99

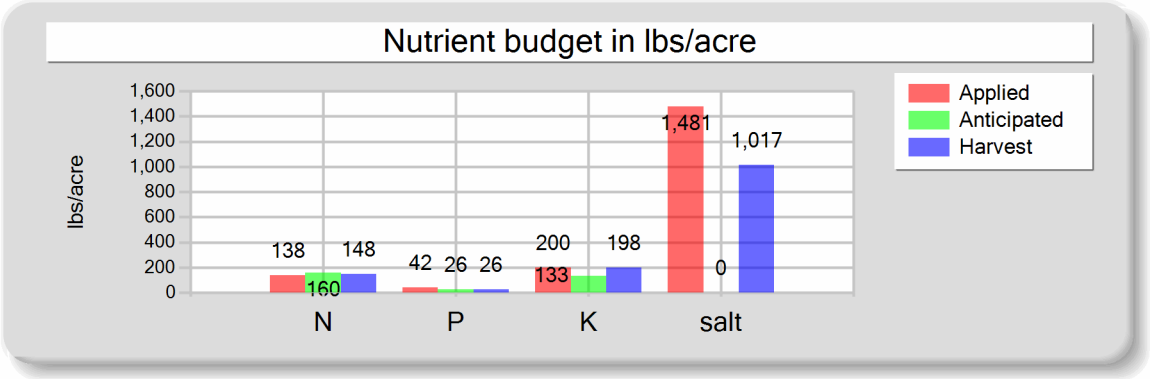
Fresh water applied	
41,368,860.00	*gallons*
1,523.47	*acre-inches*
41.17	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

D2-2 - 11/23/2022: Oats, silage-soft dough

Field name: D2-2 Crop: Oats, silage-soft dough Plant date: 11/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	120.09	42.29	200.01	1,403.89
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	11.25	0.00	0.00	76.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	138.34	42.29	200.01	1,480.83
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	148.27	25.97	198.06	1,017.01
Nutrient balance	-9.93	16.31	1.95	463.81
Applied to removed ratio	0.93	1.63	1.01	1.46

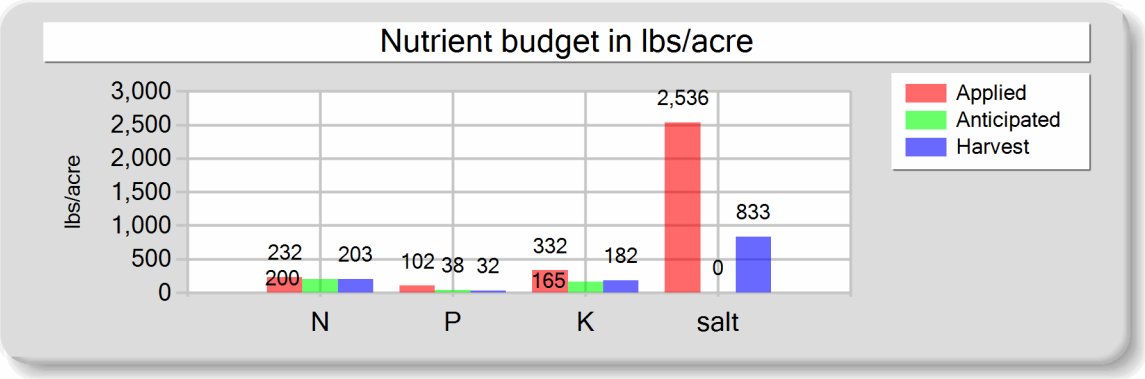
Fresh water applied	
1,613,520.00	*gallons*
59.42	*acre-inches*
4.24	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

D2-2 - 06/08/2023: Corn, silage

Field name: D2-2 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	205.42	102.03	331.78	1,743.34
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	19.40	0.00	0.00	792.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	231.81	102.03	331.78	2,535.61
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	202.51	31.56	182.00	833.18
Nutrient balance	29.30	70.47	149.78	1,702.43
Applied to removed ratio	1.14	3.23	1.82	3.04

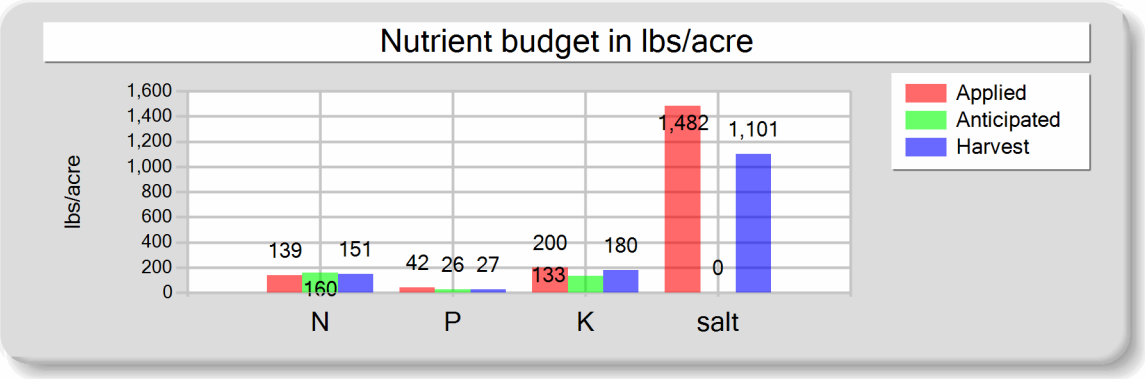
Fresh water applied	
12,669,600.00	*gallons*
466.58	*acre-inches*
33.33	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

D2-3 - 11/23/2022: Oats, silage-soft dough

Field name: D2-3 Crop: Oats, silage-soft dough Plant date: 11/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	120.09	42.29	200.01	1,403.89
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	11.46	0.00	0.00	78.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	138.55	42.29	200.01	1,482.25
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	150.63	27.16	179.64	1,101.13
Nutrient balance	-12.08	15.12	20.37	381.13
Applied to removed ratio	0.92	1.56	1.11	1.35

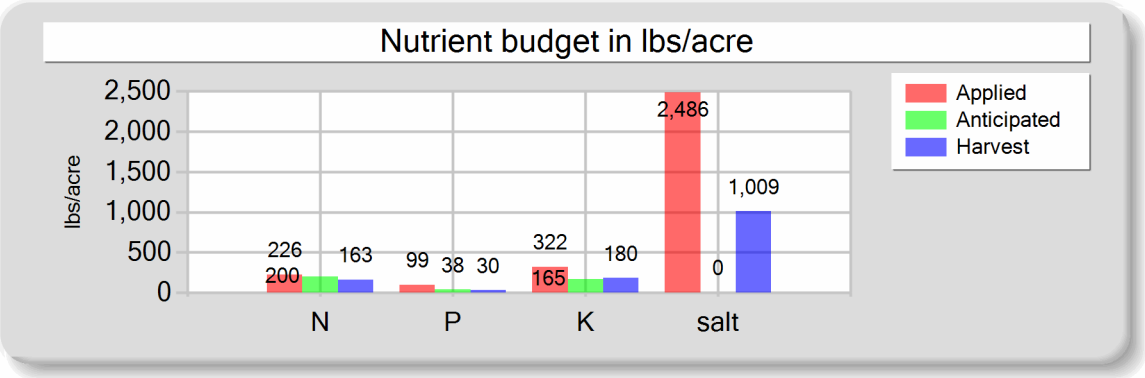
Fresh water applied	
4,930,200.00	*gallons*
181.56	*acre-inches*
4.32	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

D2-3 - 06/08/2023: Corn, silage

Field name: D2-3 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.19	98.94	321.72	1,690.51
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	19.40	0.00	0.00	795.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.59	98.94	321.72	2,486.24
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	162.87	29.90	180.49	1,008.88
Nutrient balance	62.72	69.04	141.23	1,477.36
Applied to removed ratio	1.39	3.31	1.78	2.46

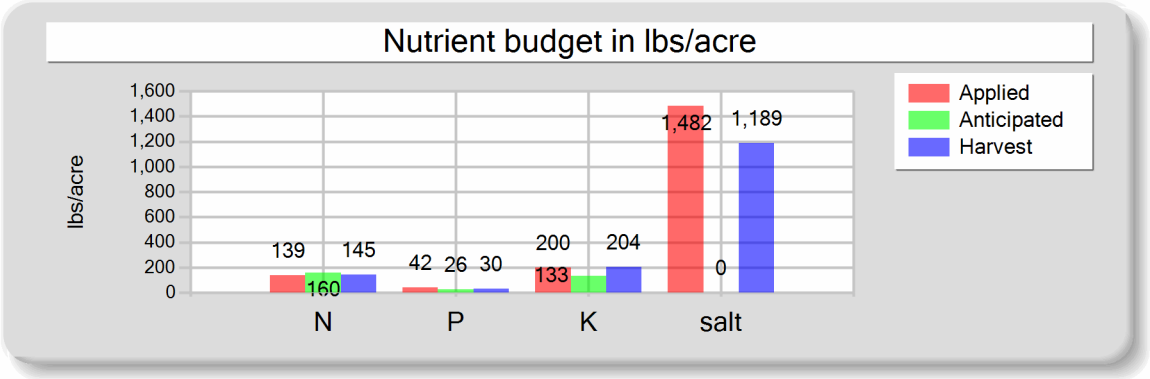
Fresh water applied	
38,308,800.00	*gallons*
1,410.78	*acre-inches*
33.59	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

D2-4 - 11/23/2022: Oats, silage-soft dough

Field name: D2-4 Crop: Oats, silage-soft dough Plant date: 11/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	120.09	42.29	200.01	1,403.89
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	11.46	0.00	0.00	78.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	138.55	42.29	200.01	1,482.25
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	144.83	30.00	204.22	1,189.06
Nutrient balance	-6.28	12.29	-4.21	293.20
Applied to removed ratio	0.96	1.41	0.98	1.25

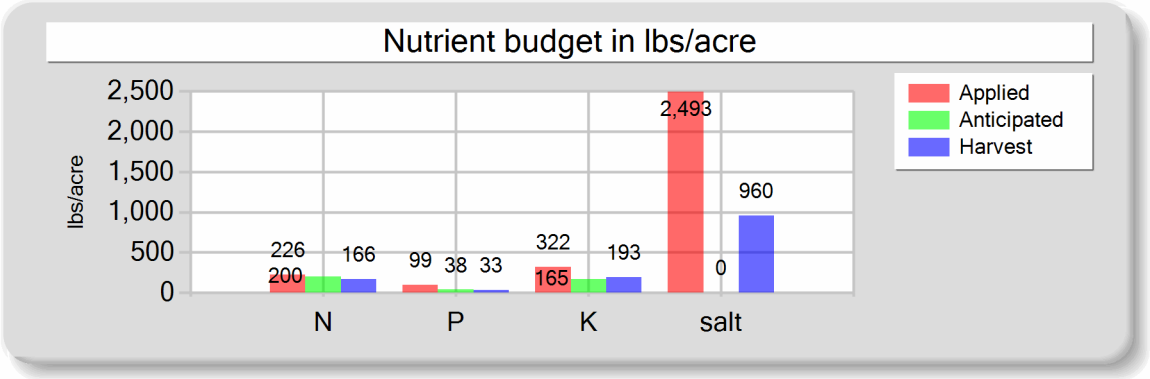
Fresh water applied	
4,930,200.00	*gallons*
181.56	*acre-inches*
4.32	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

D2-4 - 06/06/2023: Corn, silage

Field name: D2-4 Crop: Corn, silage Plant date: 06/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.19	98.94	321.72	1,690.51
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	19.83	0.00	0.00	802.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	226.02	98.94	321.72	2,492.55
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	165.98	32.76	193.28	959.93
Nutrient balance	60.04	66.18	128.44	1,532.62
Applied to removed ratio	1.36	3.02	1.66	2.60

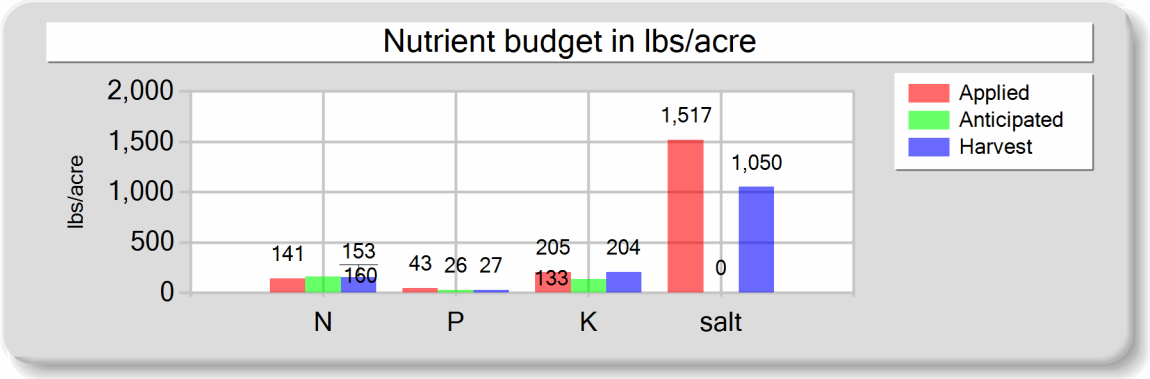
Fresh water applied	
38,173,440.00	*gallons*
1,405.80	*acre-inches*
33.47	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

D2-5 - 11/23/2022: Oats, silage-soft dough

Field name: D2-5 Crop: Oats, silage-soft dough Plant date: 11/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	123.29	43.41	205.35	1,441.32
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	11.09	0.00	0.00	75.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	141.38	43.41	205.35	1,517.12
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	153.07	26.82	204.47	1,049.95
Nutrient balance	-11.69	16.60	0.87	467.18
Applied to removed ratio	0.92	1.62	1.00	1.44

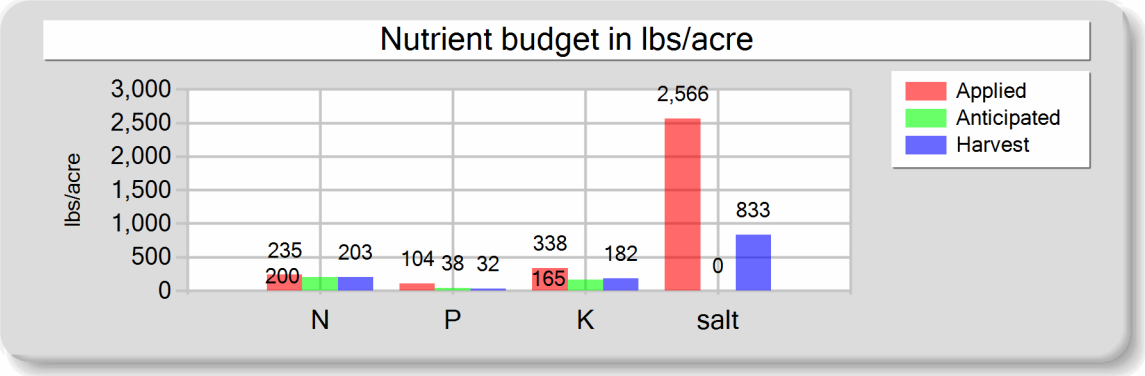
Fresh water applied
1,703,160.00 *gallons*
62.72 *acre-inches*
4.18 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

D2-5 - 06/08/2023: Corn, silage

Field name: D2-5 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	209.15	103.89	337.81	1,775.04
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	19.31	0.00	0.00	791.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.46	103.89	337.81	2,566.05
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	202.51	31.56	182.00	833.18
Nutrient balance	32.95	72.33	155.81	1,732.86
Applied to removed ratio	1.16	3.29	1.86	3.08

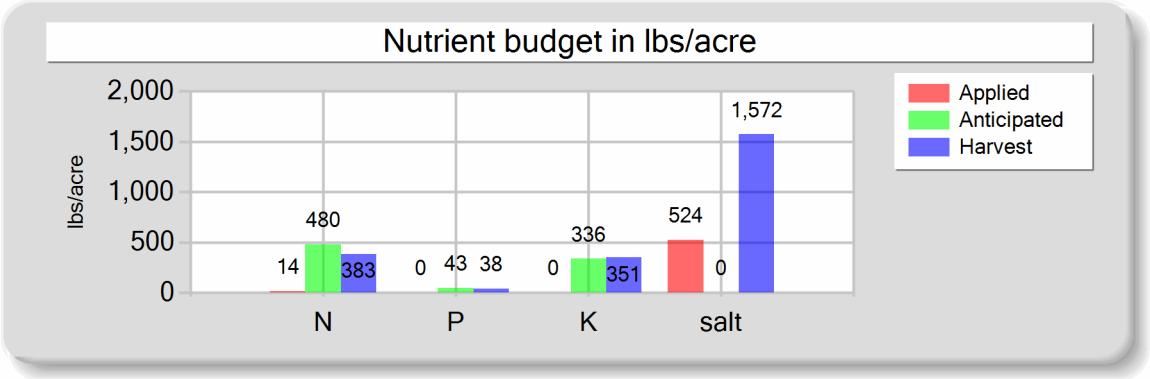
Fresh water applied	
13,584,240.00	*gallons*
500.26	*acre-inches*
33.35	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Raymus Manteca - 10/30/2019: Alfalfa, hay

Field name: Raymus Manteca Crop: Alfalfa, hay Plant date: 10/30/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	524.34
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	524.34
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	383.32	38.18	350.91	1,571.76
Nutrient balance	-369.32	-38.18	-350.91	-1,047.42
Applied to removed ratio	0.04	0.00	0.00	0.33

Fresh water applied	
29,250,000.00	*gallons*
1,077.18	*acre-inches*
39.90	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids
Sample date: 09/12/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,400.00	10,000.00	47,300.00	16,000.00	10,800.00	4,540.00	8,100.00	1,170.00		33.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids
Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 57.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,400.00	15,100.00	49,100.00	25,100.00	13,800.00	17,900.00	7,200.00	3,830.00		25.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids
Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 38.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,900.00	5,900.00	32,600.00	15,300.00	7,400.00	4,400.00	4,600.00	2,110.00		63.50
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

No process wastewater analyses entered.

C. FRESH WATER ANALYSES

Creek

Creek

Sample description: Creek

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	24.20	0.00	0.00								906.00	500
DL	0.50	0.50	0.50								1.00	10

SSJID Canal

SSJID Canal

Sample description: SSJID Canal

Sample date: 08/18/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								89.20	58
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

Brennan/120

Soil

Sample and source description: Soil

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			48.00				
DL			4.00				

D2-1

Soil

Sample and source description: Soil

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			23.00				
DL			4.00				

D2-2

Soil

Sample and source description: Soil

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			28.00				
DL			4.00				

D2-3

Soil

Sample and source description: Soil

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			39.00				
DL			4.00				

D2-4

D2-4

Soil

Sample and source description: Soil

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			68.00				
DL			4.00				

D2-5

Soil

Sample and source description: Soil

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			28.00				
DL			4.00				

Raymus Manteca

Soil

Sample and source description: Soil

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			46.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Brennan/120 - 11/22/2022: Oats, silage-soft dough

Brennan/120 - 11/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,860.00	670.00	4,730.00		8.65
DL	100.00	100.00	100.00		0.01

Brennan/120 - 06/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,740.00	590.00	3,230.00		4.84
DL	100.00	100.00	100.00		0.01

D2-1 - 10/20/2019: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	2,620.00	18,600.00		8.29
DL	100.00	100.00	100.00		0.01

D2-2 - 11/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,480.00	960.00	7,320.00		9.54
DL	100.00	100.00	100.00		0.01

D2-2 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,850.00	600.00	3,460.00		4.40
DL	100.00	100.00	100.00		0.01

D2-3 - 11/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,880.00	880.00	5,820.00		9.29
DL	100.00	100.00	100.00		0.01

D2-3 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,050.00	560.00	3,380.00		5.76
DL	100.00	100.00	100.00		0.01

D2-4 - 11/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,780.00	990.00	6,740.00		9.86
DL	100.00	100.00	100.00		0.01

D2-4 - 06/06/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,040.00	600.00	3,540.00		6.37
DL	100.00	100.00	100.00		0.01

D2-5 - 11/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,480.00	960.00	7,320.00		9.54
DL	100.00	100.00	100.00		0.01

D2-5 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,850.00	600.00	3,460.00		4.40
DL	100.00	100.00	100.00		0.01

Raymus Manteca - 10/30/2019: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,900.00	2,580.00	23,710.00		11.80
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

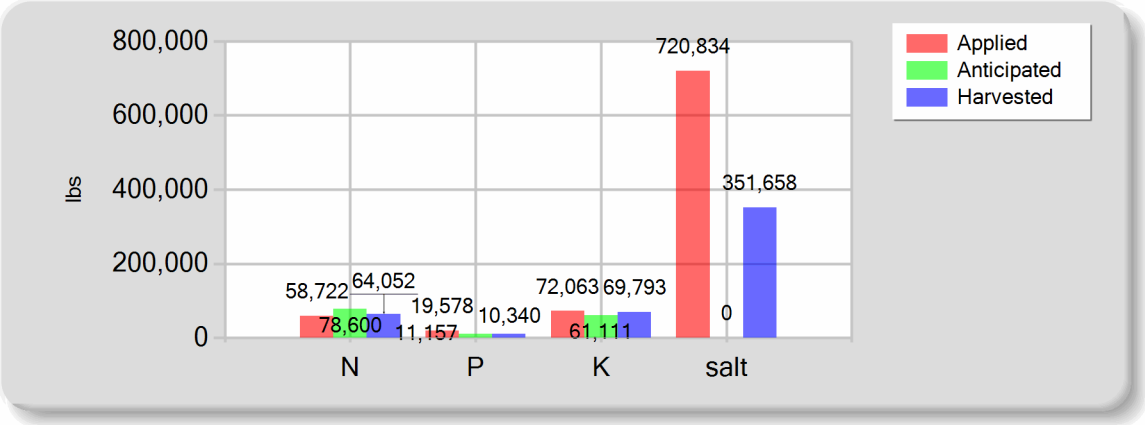
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

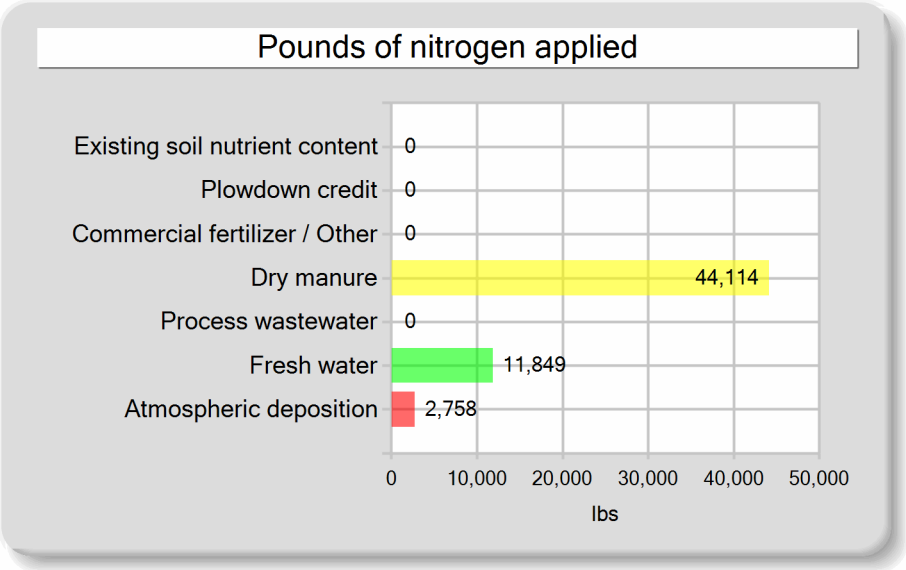
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

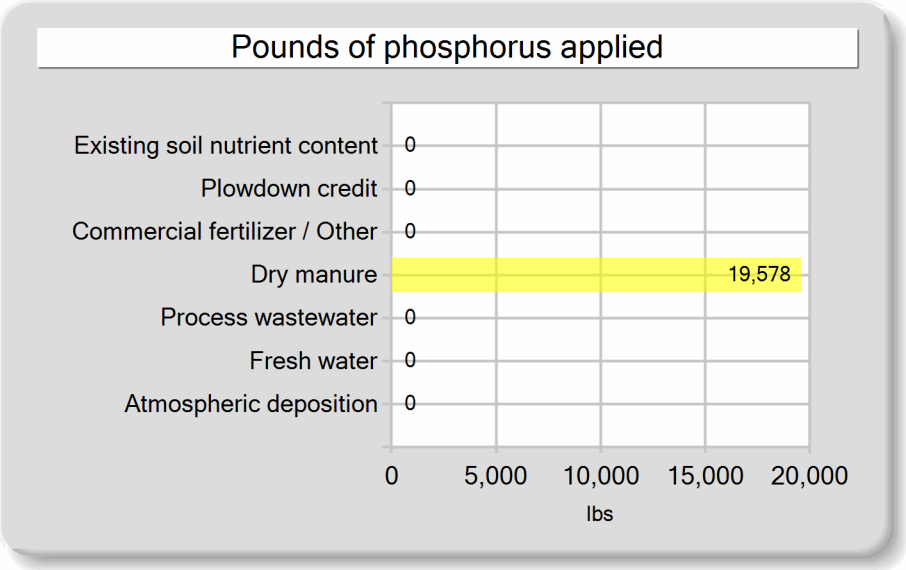
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	44,114.37	19,578.22	72,063.36	426,093.84
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	11,849.49	0.00	0.00	294,740.28
Atmospheric deposition	2,758.00	0.00	0.00	0.00
Total nutrients applied	58,721.86	19,578.22	72,063.36	720,834.12
Anticipated crop nutrient removal	78,600.00	11,157.10	61,111.40	0.00
Actual crop nutrient removal	64,051.94	10,340.02	69,793.06	351,657.74
Nutrient balance	-5,330.08	9,238.20	2,270.30	369,176.38
Applied to removed ratio	0.92	1.89	1.03	2.05

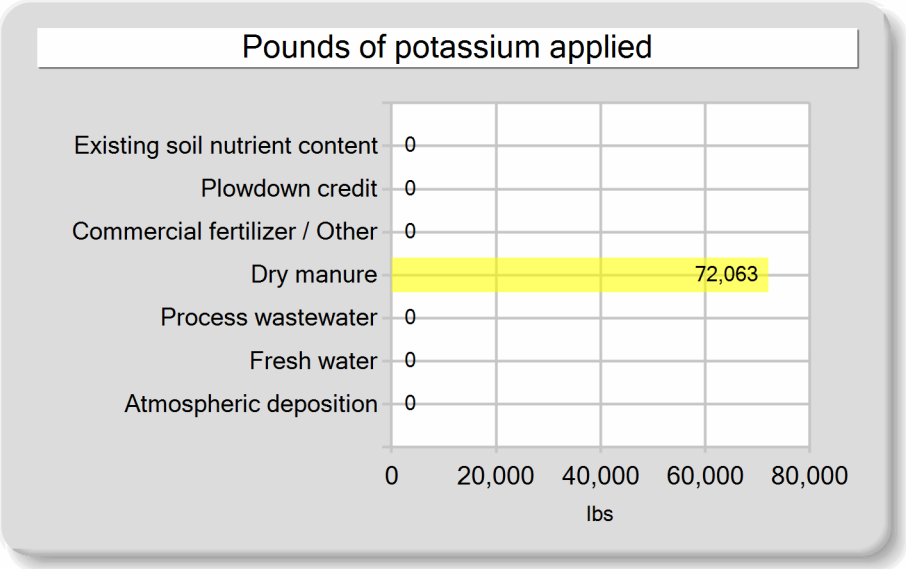
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

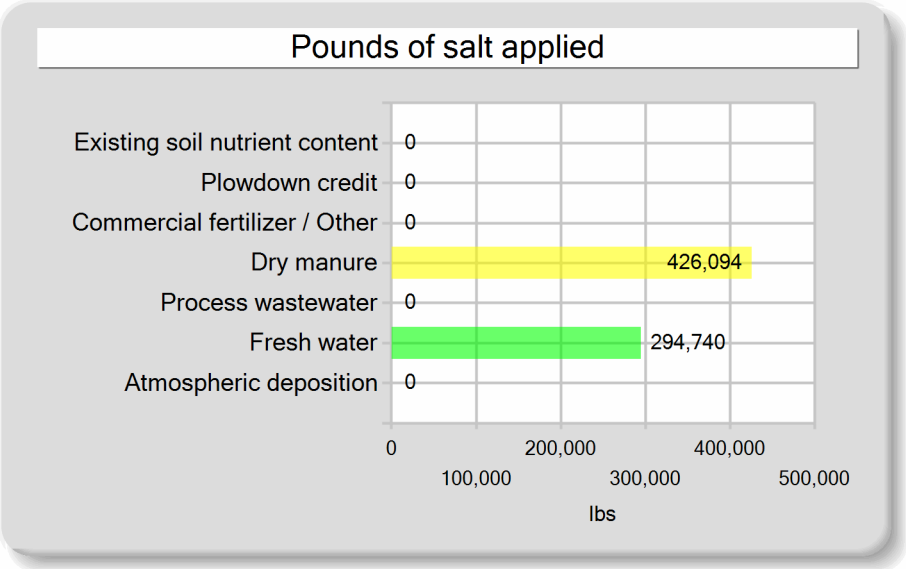


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

David Azevedo (May 8, 2024 20:34 PDT)	George Brasil (May 9, 2024 09:35 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
David Azevedo	George Brasil
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 8, 2024	May 9, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Brasil & Sons Dairy Inc
21728 Lone Tree Rd.
Escalon, CA 95220

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: #2
RECEIVED DATE: 12/15/23 11:30 AM
SUBMITTED BY: Jeanette
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 8.8 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy	T3349002A	Electrical Conductivity (EC)	EPA 120.1 and	0.904	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	22.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	579	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3349002A)	100 (T3349002A)	1.20
Electrical Conductivity (EC)	99.0			0.033
Nitrate Nitrogen (NO3-N)	105	107 (T3346026B)	105 (T3346026B)	1.40

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Brasil & Sons Dairy Inc
21728 Lone Tree Rd.
Escalon, CA 95220

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: #2
RECEIVED DATE: 12/15/23 11:30 AM
SUBMITTED BY: Jeanette
ANALYZED DATE: 12/19/23

Irrigation/Ground Water (H6)
SOURCE: Irrigation Water

NOTES: Temperature above qualifying at 11.0 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Creek	T3349002B	Total Dissolved Solids (TDS)	SM 2540 C-1997 and	500	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	24.2	mg/L
		Electrical Conductivity (EC)	EPA 120.1 and	0.906	mmhos/cm

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Electrical Conductivity (EC)	99.0			0.904 (T3349002A)	0.904 (Dup)	0.033
Nitrate Nitrogen (NO3-N)	105	107 (T3346026B)	105 (T3346026B)			1.40
Total Dissolved Solids (TDS)	105			200 (T3352010A)	205 (Dup)	2.47
Total Kjeldahl Nitrogen (TKN)	99.1			0.000 (T3345031B)	0.000 (null)	

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Brasil & Sons 2
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: George
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
____ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	12/15/23		Dairy	Domestic	8.8		✓				
2	↓		Creek	Creek	11.0						H6
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
	Jeanette Sanzer	F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC. (NO)	12-15-23 @ 11:30			T3349002

(LABORATORY USE ONLY)

2023 Annual Report Brasil & Sons Dairy #2

Final Audit Report 2024-05-09

Created:	2024-05-08
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAuNvVJQvudIK6OzNToDqOMuUepBOGhJaX

"2023 Annual Report Brasil & Sons Dairy #2" History

Document created by Vince Furtado (vfurtado@fragservices.com)
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Signer Mariana Brasil (gmmoos5@gmail.com) entered name at signing as George Brasil
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